INCH-POUND
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MILITARY SPECIFICATION SHEET

WRENCH, SPLINED SOCKET, SQUARE DRIVE, HIGH STRENGTH, THIN WALL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the wrenches described herein shall consist of this document and the latest issue of MIL-W-8982.

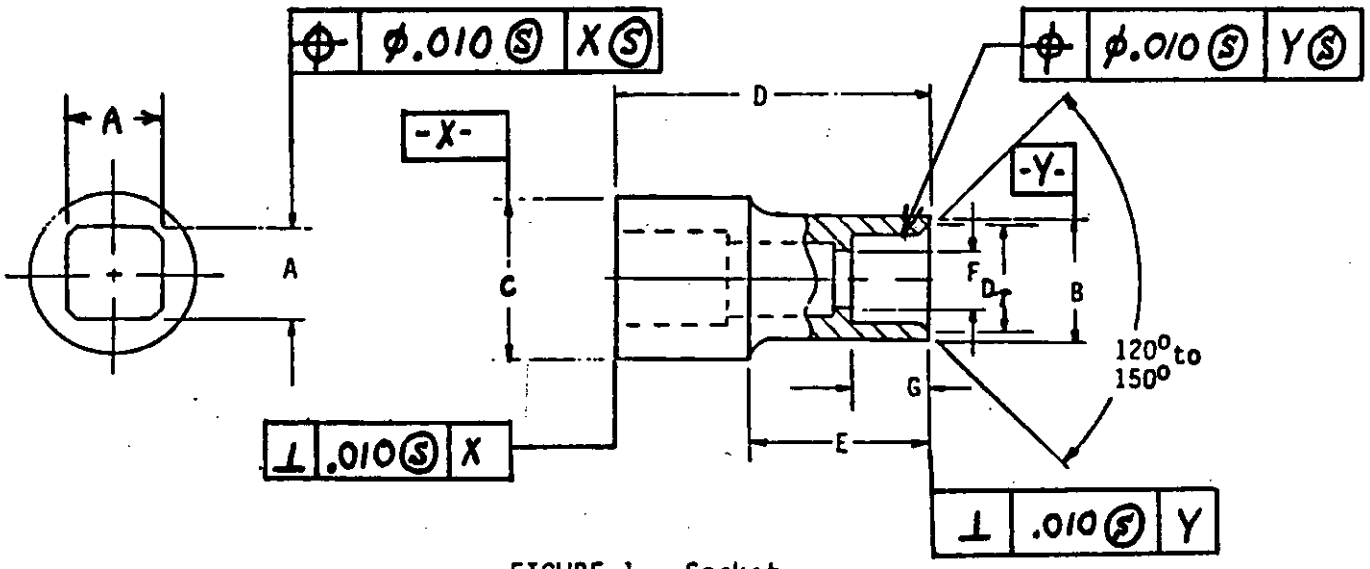


FIGURE 1. Socket.

DESIGN REQUIREMENTS:

1. Socket dimensions shall be in accordance with tables I through IV, as applicable. Spline and D_T dimensions shall conform to MS33787.

2. The female drive opening (A) shall be designed for square drive dimensions in accordance with ANSI B107.4. For socket retention, either the cross hole or recess shall be used, but must be specified in the part number callout. See Notes, p. 6, for example of part number. If recesses are specified, they shall be on all four (4) sides in the female drive end of the square drive sockets. Interior surface finish of the square drive shall be smooth and free from burrs, slivers, splits, cracks, laps and seams.

AMSC N/A FSC 5120

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3. All edges and corners shall be rounded and chamfered with sharp edges removed. The inside edges of the spline opening shall be chamfered in accordance with MS33787. Interior surface finish of the spline opening shall be smooth and free from burrs, slivers, splits, cracks, laps and seams.

4. A bolt clearance hole shall be provided from the base of the spline opening to the base of the square drive. The diameter of the bolt clearance hole shall be as specified in the applicable tables.

5. The socket shall be hardened to 45-52 HRC.

6. The material of the sockets shall be Grade E4340 (UNS G43406) per AMS 6415; Grade 4140 (UNS G41400) per AMS 6381; or Grade 6150 (UNS G61500) per AMS 6448.

7. Surface roughness shall be as specified in ANSI B107.7.

PERFORMANCE REQUIREMENTS:

1. The socket shall conform to the spline test load (endurance torque and ultimate torque) values specified in Tables I thru VI. The loads shall be applied to the square drive test plug. The square drive plugs shall comply with the dimensions of ANSI B107.4 and be of suitable strength to withstand the ultimate strength of the sockets. Measures shall be taken to prevent slippage of wrenches and mandrels during tests. There shall be no damage or permanent deformation (set) to the wrenches.

2. When engaged with spline mandrels, the socket shall be tested for conformance to the endurance torque load and ultimate torque load values specified in tables I thru VI. The load shall be held for 10 seconds. Mandrel dimensions in the engaged portion shall be in accordance with MS33787, Table I, except that C and W diameters shall be held to the minimum dimension (+ 0.002, - 0.000 inches).

3. Mandrels shall be hardened to not less than 55 HRC and smoothly finished in engagement area of wrench openings.

4. The socket shall be subjected to not less than 500 torque cycles on the spline mandrel in accordance with the endurance torque load and ultimate torque load values specified in Tables I thru VI. Engagement length shall be as specified for the ultimate tests.

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TABLE I. Socket dimensions and tests loads
1/4 inch square drive, dim A (Figure 1)
regular and long length.

Dash No.	Basic Wr. Size	B Outside Dia. at Wr. End Max. (in.)	C Outside Dia. at Sq. End Max. (in.)	D Overall length ±0.060 (in.)		E Length of Wrench End ± 0.060 (in.)		F Bolt Clrc Hole Dia Min. (in.)	G Wrench Depth Min. (in.)	Spline Test		
				Reg. Max	Long Min	Reg.	Long			Ult. Torq (in.-lb)	Endur Torq (in.-lb)	L Engmt Length (in.)
06	0.188	0.310	0.440	0.875	2.000	0.406	1.531	0.160	0.180	250	175	0.091
07	0.219	0.340	0.440	0.875	2.000	0.406	1.531	0.190	0.200	300	210	0.110
08	0.250	0.375	0.440	0.875	2.000	0.406	1.531	0.220	0.250	350	240	0.125
09	0.281	0.440	0.440	0.875	2.000	———	———	0.250	0.280	600	420	0.140
10	0.312	0.460	0.460	0.875	2.000	———	———	0.280	0.312	720	500	0.156
12	0.375	0.545	0.545	0.875	2.000	———	———	0.340	0.343	720	500	0.188
14	0.438	0.619	0.619	1.000	2.000	———	———	0.410	0.421	720	500	0.219

TABLE II. Socket dimensions and tests loads
3/8 inch square drive, dim A (Figure 1)
regular and long length.

Dash No.	Basic Wr. Size	B Outside Dia. at Wr. End Max. (in.)	C Outside Dia. at Sq. End Max. (in.)	D Overall length ±0.060 (in.)		E Length of Wrench End ± 0.060 (in.)		F Bolt Clrc Hole Dia Min. (in.)	G Wrench Depth Min. (in.)	Spline Test		
				Reg. Max	Long Min	Reg.	Long			Ult. Torq (in.-lb)	Endur Torq (in.-lb)	L Engmt Length (in.)
12	0.375	0.545	0.665	1.000	2.125	0.437	1.565	0.340	0.343	1,470	1,030	0.188
14	0.438	0.635	0.690	1.063	2.125	0.500	1.565	0.410	0.421	2.500	1,750	0.219
16	0.500	0.725	0.725	1.063	2.125	———	———	0.480	0.437	2,500	1,750	0.250
18	0.562	0.825	0.825	1.063	2.375	———	———	0.530	0.453	2,500	1,750	0.281
20	0.625	0.892	0.892	1.188	2.375	———	———	0.590	0.531	2.500	1.750	0.312

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TABLE III. Socket dimensions and tests loads 1/2 inch square drive, dim A (Figure 1) regular and long length.

Dash No.	Basic Wr. Size	B Outside Dia. at Wr. End Max. (in.)	C Outside Dia. at Sq. End Max. (in.)	D Overall length ±0.060 (in.)		E Length of Wrench End ± 0.060 (in.)		F Bolt Clrc Hole Dia Min. (in.)	G Wrench Depth Min. (in.)	Spline Test		
				Reg. Max	Long Min	Reg.	Long			Ult. Torq (in.-lb)	Endur Torq (in.-lb)	L Engmt Length (in.)
16	0.500	0.725	0.875	1.250	3.250	0.625	2.625	0.480	0.437	4,500	2,400	0.250
18	0.562	0.825	0.875	1.250	3.250	0.625	2.625	0.530	0.453	6,500	3,500	0.281
20	0.625	0.892	0.892	1.500	3.250	———	———	0.590	0.531	7,000	4,900	0.312
22	0.688	0.972	0.972	1.500	3.250	———	———	0.660	0.562	7,000	4,900	0.344
24	0.750	1.041	1.041	1.500	3.250	———	———	0.720	0.562	7,000	4,900	0.375
26	0.812	1.122	1.122	1.500	3.250	———	———	0.780	0.562	7,000	4,900	0.406
28	0.875	1.182	1.182	1.500	3.250	———	———	0.840	0.562	7,000	4,900	0.438
30	0.938	1.271	1.271	1.500	3.250	———	———	0.900	0.562	7,000	4,900	0.469

TABLE IV. Socket dimensions and tests loads 3/4 inch square drive, dim A (Figure 1) regular and long length.

Dash No.	Basic Wr. Size	B Outside Dia. at Wr. End Max. (in.)	C Outside Dia. at Sq. End Max. (in.)	D Overall length ±0.060 (in.)		E Length of Wrench End ± 0.060 (in.)		F Bolt Clrc Hole Dia Min. (in.)	G Wrench Depth Min. (in.)	Spline Test		
				Reg. Max	Long	Reg.	Long			Ult. Torq (in.-lb)	Endur Torq (in.-lb)	L Engmt Length (in.)
24	0.750	1.050	1.375	1.750		0.750		0.720	0.625	9,000	6,300	0.375
26	0.812	1.160	1.375	1.750		0.750		0.780	0.656	12,600	8,800	0.406
28	0.875	1.265	1.375	1.812		0.875		0.840	0.718	14,300	10,000	0.438
30	0.938	1.375	1.375	1.875		———		0.900	0.781	17,600	12,300	0.469
32	1.000	1.460	1.460	1.938		———		0.960	0.812	20,000	14,000	0.500
34	1.062	1.545	1.545	2.000		———		1.020	0.875	20,000	14,000	0.531

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TABLE V. Socket dimensions and tests loads 1 inch square drive, dim A (Figure 1) regular and long length.

Dash No.	Basic Wr. Size	B Outside Dia. at Wr. End Max. (in.)	C Outside Dia. at Sq. End Max. (in.)	D Overall length ±0.060 (in.)		E Length of Wrench End ± 0.060 (in.)		F Bolt Clrc Hole Dia Min. (in.)	G Wrench Depth Min. (in.)	Spline Test		
				Reg. Max	Long	Reg.	Long			Ult. Torq (in.-lb)	Endur Torq (in.-lb)	L Engmt Length (in.)
32	1.000	1.460	2.000	2.438		1.000		.960	0.812	22,000	15,400	0.500
34	1.062	1.545	2.000	2.500		1.190		1.020	0.875	26,350	18,450	0.531
36	1.125	1.630	2.000	2.562		1.250		1.080	0.937	31,300	21,950	0.563
38	1.188	1.715	2.000	2.750		1.440		1.200	1.140	36,800	25.750	0.594
40	1.250	1.795	2.000	2.750		1.440		1.200	1.062	42,950	30,100	0.625
42	1.312	1.890	2.000	2.875		1.560		1.270	1.125	47,000	32,900	0.656
44	1.375	1.980	2.000	2.938		1.620		1.340	1.188	47,000	32,900	0.688
46	1.438	2.070	2.070	3.000		———		1.400	1.250	47,000	33,900	0.719
48	1.500	2.155	2.155	3.062		———		1.470	1.312	47,000	32,900	0.750

TABLE VI. Socket dimensions and tests loads 1-1/2 inch square drive, dim A (Figure 1) regular and long length.

Dash No.	Basic Wr. Size	B Outside Dia. at Wr. End Max. (in.)	C Outside Dia. at Sq. End Max. (in.)	D Overall length ±0.060 (in.)		E Length of Wrench End ± 0.060 (in.)		F Bolt Clrc Hole Dia Min. (in.)	G Wrench Depth Min. (in.)	Spline Test		
				Reg. Max	Long	Reg.	Long			Ult. Torq (in.-lb)	Endur Torq (in.-lb)	L Engmt Length (in.)
42	1.312	1.890	3.000	3.250		1.560		1.270	1.125	49,700	34,800	0.656
44	1.375	1.980	3.000	3.312		1.650		1.340	1.188	57,200	40,050	0.688
46	1.438	2.070	3.000	3.375		1.690		1.400	1.250	65,400	45,800	0.719
48	1.500	2.155	3.000	3.438		1.750		1.470	1.312	74,250	52,000	0.750
50	1.562	2.240	3.000	3.468		1.780		1.530	1.343	83,850	58,700	0.781
52	1.625	2.325	3.000	3.500		1.810		1.600	1.375	94,400	66,100	0.813
54	1.688	2.325	3.000	3.562		1.870		1.660	1.438	105,800	74,050	0.844
56	1.750	2.495	3.000	3.625		1.940		1.720	1.500	117,900	82,550	0.875
58	1.812	2.580	3.000	3.687		2.000		1.790	1.562	130,900	91,600	0.906
60	1.875	2.665	3.000	3.750		2.060		1.850	1.625	145,000	101,500	0.938
62	1.938	2.755	3.000	3.812		2.120		1.870	1.656	155,000	108,500	0.969
64	2.000	2.840	3.000	3.844		2.160		1.930	1.688	170,000	119,000	1.000
66	2.062	2.925	3.000	3.875		2.190		1.990	1.718	180,000	126,000	1.031
68	2.125	3.010	3.010	3.906		———		2.060	1.750	190,000	133,000	1.062
70	2.188	3.100	3.100	3.938		———		2.120	1.781	200,000	140,000	1.094

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NOTES:

1. Dimensions are in inches.

2. Sockets shall be chromium plated in accordance with QQ-C-320, Class 1, Type I, unless otherwise specified. Cadmium plating is not permitted.

3. The sockets shall be of uniform quality. They shall be free from nicks, splits, cracks, laps, seams and scale. They shall have no imperfections that will adversely affect serviceability, reliability or safety.

4. Example of a military part number:

M8982/1 - 06 2 R A

- A = Recess; B = Cross hole
- R = Regular length; L = Long length
- Square drive number in eighths of an inch (1/4=2, 3/8=3, 1/2=4, 3/4=6, 1=8, 1 1/2=12)
- Dash number in thirty seconds of an inch
- Basic specification sheet number

5. For design purposes , this specification sheet takes precedence over procurement documents referenced herein.

Custodians:
Army - AV
Air Force - 99
Navy - AS

Review:
Army - GL, AR

Preparing Activity:
Navy - AS

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